

CORE_RULE: Python Computes. LLM Narrates.

FinVeritas V2: From Explainable Analysis to Explainable Forecasting

Project Review & Architectural Evolution

Target Domain: SaaS & Listed Enterprises

Validation Focus: TCS, Infosys, HDFC Bank

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Evolving the Platform: From Historical Audits to Predictive Intelligence

	V1	V2
Core Question	Answers: What happened?	Answers: What will happen, why, and how confident should you be?
Scope	Single-company analysis.	Competitive, market-aware, and governance-screened.
Agent Pipeline	5-agent historical pipeline.	13-capability intelligence pipeline.
Risk Monitoring	One-time Credibility Engine check.	Ongoing continuous Anomaly Alert System.
Calculations	Computes a handful of targeted ratios.	Foundational 12-category Deterministic Financial Metrics Engine.

The Empirical Divide: Why Arithmetic Must Remain Outside the LLM

Diagnostic Justification Matrix		
Research Paper	Identified LLM Failure Mode	FinVeritas V2 Architectural Solution
FAITH (2025)	95.6% -> ~0% accuracy collapse. General LLMs fail completely on multivariate financial calculations.	Strict isolation. 100% of arithmetic is handled in Pandas/NumPy.
FinVerBench (2026)	Predictive accuracy drops to random chance when financial statements are perturbed, proving LLMs memorize rather than reason.	Implementation of deterministic fact ledgers before LLM narration.
VC Due-Diligence Orchestration (2026) & Macro Economists in the Machine (2026)	Mega-prompts fail at domain vocabulary and precision.	Role-based multi-agent orchestration (specialized agents for Geography, Competitors, Credit).

Target Domain & Validation: Listed SaaS and IT Enterprises

Domain Rationale

- Standardized and highly transparent financial reporting structures.
- Heavy emphasis on recurring revenue (ARR/MRR) metrics, ideal for deterministic trend analysis.
- Consistent geographic segment reporting, enabling granular spatial analysis.

Feasibility Validation Subjects

COMPANY_01: Tata Consultancy Services (TCS)

COMPANY_02: Infosys

COMPARATIVE: HDFC Bank (Banking Parallels)

SOURCE: Bloomberg Standardized Financial Statement Analysis Reports

The Deterministic Financial Metrics Engine: 90% Feasibility Achieved

Feasibility Matrix

100% Python Computable (From Ingested Data)

Income Statement Metrics (100%)

Balance Sheet Metrics (100%)

Cash Flow Metrics (100%)

Profitability & Return Ratios (100%)

Liquidity & Solvency Ratios (100%)

Growth & Trend Intelligence (100%)

Efficiency & Risk Monitoring (~95%)

The 10% External Gap (Next Phase Integrations)

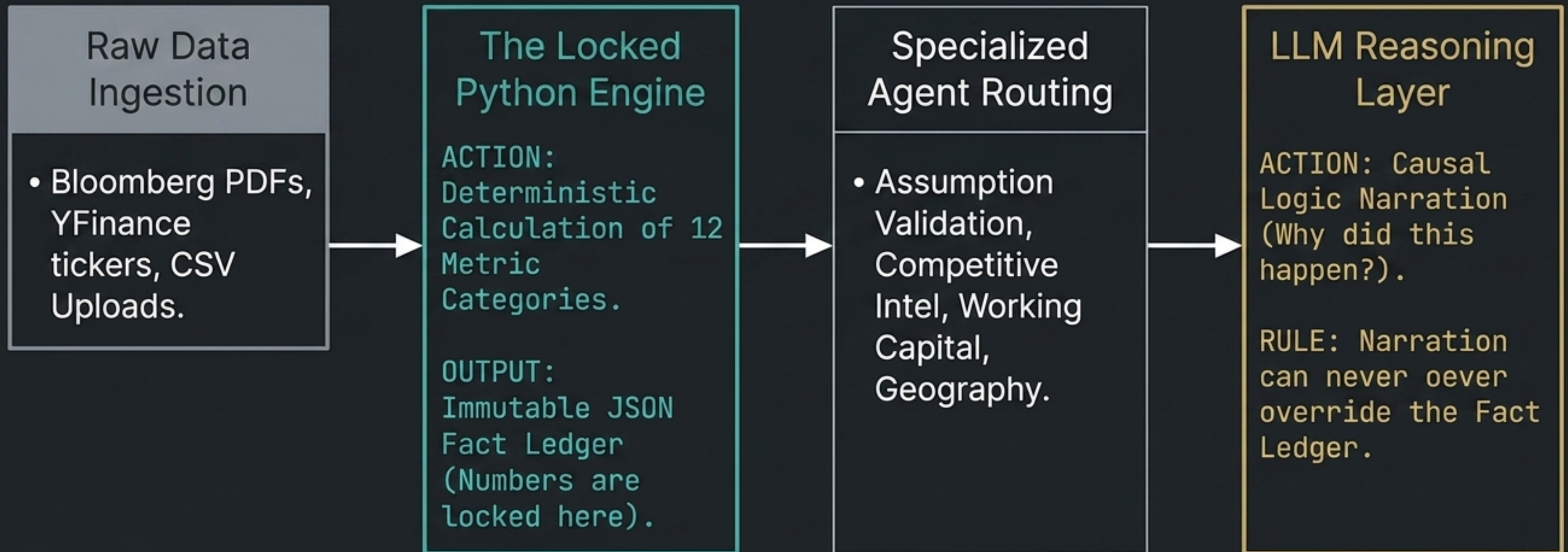
Market Valuation (P/E, EV/EBITDA) -> Requires market data feeds.

Precise DSCR -> Requires debt maturity schedules.

Governance Intelligence -> Requires external statutory data.

Competitive Benchmarking -> Requires peer financials.

The Immutable Pipeline: Bridging Computation and Narration



Expanding Scope: Continuous Risk and Competitive Intelligence

Anomaly Alert System (Self-Awareness)

- Continuously monitors structural ratios (Debt/Equity, Net Asset/Total Debt) rather than relying on one-time checks.

TRIGGER EXAMPLE:
IF [Net Asset / Total Debt] drops
below historical sector norm
-> TRIGGER WARNING

Competitive Intelligence Agent (Competitive Edge)

- Positions the target company against direct peers utilizing external market data feeds.

TRIGGER EXAMPLE:
COMPARE [Company EBITDA Margin] vs
[Sector Median] -> NARRATE GAP

Intelligence Pipeline I: Assumption Validation & Forecasting

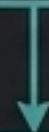
Step 1: User Input

Human inputs a projected growth assumption (e.g., 15% ARR Growth).



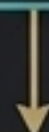
Step 2: Python Extraction

System pulls management guidance from earnings calls, analyst estimates, and historical CAGR.



Step 3: Deterministic Cross-Check

Python calculates the delta between the user assumption and the sector benchmark.



Step 4: LLM Narration

The LLM flags significant deviations and narrates the rationale (e.g., 'Management guided 8-10%, user assumption of 15% is aggressive given sector headwinds').

[Regional Revenue Volatility %]

Deterministically computed from annual report segment disclosures via YoY comparison.

+

[Lexicon Event Flag]

Geopolitical incidents, natural disasters, currency moves sourced from Sentiment Agent news pipeline.

=

[Explainable Narrative]

APAC revenue declined 12% YoY, correlating with documented supply chain disruptions in the region.

CORE RULE REINFORCEMENT: The LLM explains the regional swing; the underlying revenue figures remain immutable.

Intelligence Pipeline II: Bank-Grade Working Capital Assessment

1.	INPUT:	Net Sales & Total Current Assets (TCA)
2.	INPUT:	Current Liabilities (excluding bank borrowings)
3.	COMPUTE:	Working Capital Gap (WCG) = TCA - Current Liabilities
4.	COMPUTE:	Minimum Stipulated Net Working Capital (NWC) = 25% of TCA
5.	OUTPUT:	Maximum Permissible Bank Finance (MPBF) = WCG - NWC

Extends FinVeritas from historical analysis into preliminary, explainable credit appraisal via the Tandon Method (Method II).

Intelligence Pipeline II: Debt Servicing Capability (DSCR)

-> Represents Cash Flow Available for Debt Service.

$$\text{DSCR} = \frac{(\text{Profit After Tax [PAT]} + \text{Depreciation} + \text{Interest})}{(\text{Principal Repayment} + \text{Interest})}$$

-> Represents Total Debt Obligation.

DATA REQUIREMENT NOTE: Precise calculation requires integration of external Debt Maturity Schedules (flagged in the 10% feasibility gap).

The Governance Pivot: Executive Background & Statutory Trust

Step 1: DIN Extraction

Extract Director Identification Number (DIN) from uploaded filings.



Step 2: Deterministic Statutory Retrieval

- Query **Ministry of Corporate Affairs (MCA)** for active/disqualified status.
- Query **SEBI** (Securities and Exchange Board of India) for debarment lists.
- Query **IBBI** (Insolvency and Bankruptcy Board) for prior company failures.



Step 3: Governance Agent Output

Screen leadership history for red flags and conflicts of interest, narrating the findings alongside the financial credibility score.

Because the LLM never touches the raw financial data or performs calculations, FinVeritas inherently achieves a “verified data chain.”

DPDP Alignment

Deterministic data extraction and DIN-only resolution ensure strict data privacy and purpose limitation. No sensitive data is hallucinated or leaked by the LLM.

SOC Readiness

The immutable **Python** fact-ledger provides a **100%** auditable trail for every metric, a mandatory requirement for SOC compliance.

No-Judgment Reporting

The **LLM** provides explainability, but the financial health classification (**PASS/WARN/FAIL**) is driven purely by deterministic math, removing AI bias from the equation.

Future Roadmap: Closing the 10% Intelligence Gap

